

Workshop: Sampling Toolkit in PyTorch



Goal: sampling is a *procedure*—we start from uniform randomness and transform it into samples from more complex distributions.

You will implement and visualise:

- ① **Categorical sampling (given probability mass functions):** sample an index $k \sim \text{Cat}(\pi)$.
- ② **Inverse CDF sampling (1D):** Exponential

$$u \sim \mathcal{U}(0, 1), \quad x = -\frac{1}{\lambda} \log(1 - u).$$

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3 Known transform (Gaussian): Box–Muller

$$u_1, u_2 \sim \mathcal{U}(0, 1) \Rightarrow \begin{cases} z_1 = \sqrt{-2 \log u_1} \cos(2\pi u_2), \\ z_2 = \sqrt{-2 \log u_1} \sin(2\pi u_2). \end{cases}$$

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- ④ **multimodal**: 2D mixture of Gaussians
- **Pick which mode/component to use:**

$$k \sim \text{Cat}(\pi).$$

Here $k \in \{1, \dots, K\}$ is an index (which cluster you'll sample from), and $\pi = (\pi_1, \dots, \pi_K)$ is a PMF over clusters (mixture weights). If π is uniform, all clusters are equally likely; if π is imbalanced, some clusters are sampled more often.

- **Sample a 2D point from that Gaussian:**

$$x \sim \mathcal{N}(\mu_k, \sigma^2 I).$$

Here $x \in \mathbb{R}^2$ is the sampled point, μ_k is the center of cluster k . Overall, the distribution of x is

$$p(x) = \sum_{k=1}^K \pi_k \mathcal{N}(x; \mu_k, \sigma^2 I),$$

which is *multimodal* (multiple peaks).



What you should observe (and why it matters)

Expected visual outputs

- **Categorical:** empirical histogram $\approx \pi$ (law of large numbers).
- **Exponential (inverse CDF):** histogram concentrated near 0 with a long tail.
- **2D Gaussian (Box–Muller):** circular “blob” centered at the origin.
- **Mixture (8 modes):** 8 clusters on a ring (multimodality).

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Quick diagnostic: mode coverage

assign each sample to nearest center $\mu_k \Rightarrow$ how many modes received any samples?

- With uniform π , frequencies should be roughly balanced.
- With imbalanced π , the sample frequencies track π (some modes become rare).



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Key takeaway: for simple distributions, sampling is a closed-form transform; for richer data, we need *structured sampling procedures* (sequential, latent, iterative) to generate complex, multimodal x .